

Application of Hybrid Particle Swarm Optimization Algorithm Based on Entropy in Flexible Job Shop Scheduling

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Abstract: In order to better solve flexible shop scheduling problems of various scales, a hybrid particle swarm optimization algorithm based on entropy is designed by combining particle swarm optimization with simulated annealing algorithm and adaptively adjusting inertia coefficient according to population entropy. The simulation results prove that this algorithm can effectively deal with flexible operation shop scheduling problem, and it has obvious advantages compared with the traditional optimization algorithm.

Keywords: information entropy, simulated annealing algorithm, particle swarm optimization, flexible job shop scheduling

1. Introduction

Particle swarm optimization, or particle swarm optimizer, PSO algorithm, is a swarm intelligence-based intelligent optimization algorithm proposed by Drs. Eberhart and Kennedy [1]. It solves the problem by simulating the foraging behavior of birds and generates a random set of initial solutions. Each individual flies at a particular speed in the search area, adjusting based on his own and his companions' flying experience. PSO is widely used in function improvement, fuzzy system regulation, neural network learning and other categories and other application fields due to its few individuals, straightforward calculation, and high robustness. Information entropy is a mathematical abstraction, so we might as well interpret it as the probability of a certain piece of information occurring (the probability of discrete random events occurring). The information entropy of a system is lower the more organized it is. Inversely, the greater a system's chaos, the greater its information entropy. The entropy of information is a measure of the degree of system ordering.

FJSP is a further extension of the traditional work-shop allocation problem [2]. In addition to being an NP-Hard problem, it has a larger feasible solution space and greater complexity than the classic job shop problem. Although many algorithms have been used to deal with these kinds of allocation problems, for example, reference [3] applied genetic arithmetic method to solve flexible job shop allocation problem,

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reference [4] applied particle swarm improvement arithmetic method to solve flexible job shop allocation problem, and reference[5] use imperialist competitive algorithm for solving the flexible job shop scheduling problem, the effect of a single algorithm is not optimal. This paper combines particle swarm optimization with simulated annealing, modifies the adaptive inertia coefficient based on population entropy, and at the same time, a kind of entropy-dependent composite particle swarm improvement method is presented to improve the flexible workshop allocation problem.

2. Problem Description

The flexible operation shop allocation problem is a further expansion of the traditional operation shop allocation problem, which not only inherits all the characteristics of job shop problem, but also increases the new difficulty of solving it. The flexible operation shop allocation problem is described as: parts with various production sequences are produced on (equipment, where each program can be produced in multiple equipment, and each machine can also perform processing tasks of multiple processes. It is known that workpiece sets $\mathbf{P} = \{p_1, p_2, \dots, p_n\}$ and p_i are the i th workpiece, $i = 1, 2, \dots, n$; Machine set $\mathbf{M} = \{m_1, m_2, \dots, m_n\}$, m_j is the j machine, and $j = 1, 2, \dots, m$, the machines with the same processing capacity and can complete the same process in the machine set are called parallel machines; Operation sequence set $\mathbf{OP} = \{op_1, op_2, \dots, op_n\}$, and $op_i = \{op_{i1}, op_{i2}, \dots, op_{in}\}$ is the operation sequence of workpiece p_i . Unlike the traditional Job-Shop deployment problem, the number of jobs is not necessarily equal to the number of machines, that is, there may be parallel machines or multifunctional machines that can process two or more jobs in the workshop; The time matrix $\mathbf{T}$ of each workpiece using each machine, and $t_{ij} \in \mathbf{T}$ is the time when the i workpiece p_i uses the j machine. $t_{ij} = 0$ indicates that the workpiece p_i does not use a machine j . In addition, the flexible Job Shop scheduling problem also needs to meet the following conditions: each job can be used more than once on each machine; The sequence of using each machine for each workpiece can be different, that is, there can be $op_i \neq op_d$ ($i \neq d$); Each workpiece has the same processing priority; No new workpiece is added during the processing of each workpiece, and the processing cannot be temporarily interrupted.

3. Particle Swarm Optimization Algorithm for Flexible Job Shop Scheduling

The intelligent evolutionary algorithm Particle Swarm Optimization (PSO) was proposed by Kennedy and Elberhart in 1995. The improved particle swarm computing method mainly uses the concepts of "group" and "evolution", and calculates according to the applicability value of the particle. Each particle flies at the corresponding speed in the one-dimensional retrieval space, and the flight speed is adjusted in real time by

the group flight experience and the individual flight experience. Let the scales of population S and $\mathbf{X}_i = (x_{i1}, x_{i2} \dots x_{in})$ be the current position of particle i ; $\mathbf{V}_i = (v_{i1}, v_{i2} \dots v_{in})$ is the current velocity of particle i ; $\mathbf{P}_i = (p_{i1}, p_{i2} \dots p_{in})$ is the historical best position of particle i , which is called the individual best position. The historical optimal site in the minimization problem is the site with the smallest value of the applicability function, and the optimal site passed by all particles in the population is $\mathbf{P}_g(t)$, which is called the global optimal site. The evolution equation of the basic particle swarm improvement method is shown as follows:

$$v_{ij}(t+1) = wv_{ij}(t) + c_1r_1(t)(p_{ij}(t) - x_{ij}(t)) + c_2r_2(t)(p_{gj}(t) - x_{ij}(t)) \quad (1)$$

$$x_{ij}(t+1) = x_{ij}(t) + v_{ij}(t+1) \quad (2)$$

In the formula, $v_{ij}(t), x_{ij}(t), v_{ij}(t+1), x_{ij}(t+1)$ represent the flying speed and position of the i -th particle in the t and $t+1$ generations respectively. The subscript " j " indicates the j dimension of the particle, " i " indicates the particle i , t indicates the t th generation, w is the inertia coefficient, and c_1 and c_2 are the acceleration constant of individual particles and group particles, usually taking the value of $0 \sim 2$. r_1, r_2 is a random number in the range of $(0,1)$. From the rate update expression of equation (1), it can be found that c_1 adjusts the step size of the particle in the historically optimal direction, and c_2 adjusts the step size of the particle in the globally optimal direction. To reduce the probability of particles leaving the retrieval space during evolution, v_{ij} is usually limited in the interval $[-v_{\max}, v_{\max}]$. If the search space of the problem is limited in $[-x_{\max}, x_{\max}]$, $v_{\max} = k \times x_{\max}$, $0.1 < k < 1.0$ can be set.

3.1 Encoding and Decoding of Particles

In the particle swarm improvement method for solving the flexible operation shop allocation improvement problem, each particle corresponds to a scheduling scheme. The location vector of particles is represented by a vector with dimension

$2L$ ($L = \sum_{j=1}^n n_j$ is the total number) in the problem of flexible job shop scheduling. In

fact, this position vector consists of two L -dimensional vectors, namely, the process vector $\mathbf{X}_{\text{process}}[L]$ and the machine vector $\mathbf{X}_{\text{process}}[L]$. The process vector $\mathbf{X}_{\text{process}}[L]$ consists of L non-zero natural numbers that are not greater than n (the number of workpieces), and the order of the natural numbers determines the processing order of workpiece processes. Here, the operation-based coding method in

genetic algorithm is used for reference, that is, each process vector consists of L bit representing the operation, which is an arrangement of all operations. For example, if the operation vector $\mathbf{X}_{\text{process}}[L]$ is [2 1 1 2 2 3 1 3 3], Then 1 represents part 1, 2 represents part 2, 3 represents part 3. The first generation of 1 represents part 1 as the first operating procedure, the second generation of 1 represents the second operating procedure of part 1, and the third generation of 1 represents the third operating procedure of part 1, and so on. The machine vector $\mathbf{X}_{\text{process}}[L]$ consists of L non-zero natural numbers which are not more than m (number of machines), and represents the machine number of the working procedure at the corresponding position of machining $\mathbf{X}_{\text{process}}[L]$.

3.2 Calculation of Position Vector and Velocity Vector

Particle swarm improvement (PSO) is a kind of improved algorithm in continuous space, and the flexible operation workshop allocation problem is a discrete non-numerical improvement problem; therefore, the iterative calculation process is modified as follows:

(1) For $\mathbf{X}_{\text{machine}}[L]$, because the machine vector's value is an integer in the interval $[1, m]$, if the result calculated according to formula (2) is a decimal, it will be rounded down; if the result calculated according to formula (2) exceeds the value interval, it will be taken according to the boundary.

(2) For $\mathbf{X}_{\text{process}}[L]$, because the optimization problem of flexible shop scheduling is the sequencing problem of process sequence, iterative operation is adopted. Assume that the initial value is:

Dimension:	1	2	3	4	5	6	7	8	9
$\mathbf{X}_{\text{process}}[L]$:	1	1	1	2	2	2	3	3	3 (initial value)

After an iterative operation, the following results are obtained:

$$\mathbf{X}_{\text{process}}[L] : 0.12 \quad 0.25 \quad 0.60 \quad 0.18 \quad 2.52 \quad 1.25 \quad 0.82 \quad 2.45 \quad 1.54$$

(calculated according to formula 2)

$$\text{Initial value:} \quad 1 \quad 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 3 \quad 3 \quad 3$$

The calculated values are arranged in order from smallest to largest, and the initial values corresponding to the calculated values are reordered based on this sorting, yielding the following results:

$$\text{Rank of calculated values: } 0.12 \quad 0.18 \quad 0.25 \quad 0.60 \quad 0.82 \quad 1.25 \quad 1.54 \quad 2.45 \quad 2.54.$$

$$\mathbf{X}_{\text{process}}[L] : \quad 1 \quad 2 \quad 1 \quad 1 \quad 3 \quad 2 \quad 3 \quad 3 \quad 2$$

Then the result after this iteration is [1 2 1 1 3 2 3 3 2], and a feasible solution can be obtained after this transformation.

4. Entropy-based Hybrid Particle Swarm Optimization Algorithm

4.1 Simulated Annealing Algorithm

The algorithm for simulated annealing [6] is based on the principle of solid annealing, which involves heating the solid to a sufficient temperature and then gradually cooling it. When the solid is heated, as the temperature rises, the internal particles will become disorderly, and the internal energy will increase correspondingly. However, when the solid is gradually cooled, the particles gradually become orderly, and can achieve dynamic equilibrium under various temperature conditions, reaching the ground state at room temperature, and the internal energy is reduced to the lowest. According to the Metropolis standard, the probability that a particle tends to equilibrium at temperature T is $e^{-\Delta E/(\lambda T)}$, where E mainly represents the internal energy at temperature T , ΔE mainly represents its corresponding change, and λ mainly represents the Boltzmann constant. By using the solid state annealing simulation combined improvement algorithm, the internal energy E is simulated into the objective function f , and the temperature T is transformed into the control parameter t , and then the simulated annealing operation method is formed to deal with the combined improvement problem. The loop of "creating new solution computing the objective function difference accepting or rejecting" is repeated for the present solution, starting with the initial solution i and the initial value t of the control parameter, and the value t is gradually attenuated. The solution of this algorithm is the approximate best solution obtained, which is a kind of heuristic random retrieval program based on Monte Carlo iterative algorithm. The annealing step is regulated by a cooling scheme, including the number of iterations L and the stop condition S for each value t of the initial value t of the control parameter and its decay factor Δt .

4.2 Hybrid Particle Swarm Optimization Algorithm

Due to the one-way flow of information between particles, Particle Swarm Optimization (PSO) has a very rapid convergence speed. After a few iterations, the characteristics of the population's particles are frequently identical. In this manner, the population lacks diversity, it is difficult to escape local optimum, and it is frequently simple to develop premature convergence. Nevertheless, the simulated annealing algorithm begins the domain search from a specific starting point at a specific initial temperature, with the further reduction of the temperature parameter, the global optimal solution of the objective function is randomly found in the solution space according to the probability hopping characteristics, i.e., it can jump out of the local optimal solution and eventually converge to the global optimal solution, but its calculation time is much longer.

The combination of particle swarm improvement and simulated annealing method is more effective for the following reasons: The integration of optimization mechanisms comes first. SA can effectively avoid falling into local minima and eventually tending to global optima by providing the search process with a probability jump that can change at any time and tends to zero; PSO achieves optimization through the guidance of flight information of global optimal particles and individual optimal particles. Combining the two algorithms with such distinct selection mechanisms is

advantageous for enriching the search behavior during the optimization process and enhancing the search capability and effectiveness in both the global and local senses. The combination of optimization operations is the second step. SA algorithm only retains a single solution at any given time, lacking redundancy and historical search information, whereas PSO inherits some information of the global optimal individual and the historical optimal particle in the population during the speed update procedure, ensuring that the particles continue to evolve in the right direction. The combination of these various optimization operations enriches the neighborhood search structure during the optimization process and improves the global space's search capability. Third, optimization behavior's complementarity. If the convergence criterion of particle swarm optimization is not well designed, PSO will frequently evolve slowly or converge prematurely; However, the optimization behavior of the algorithm SA has a strong dependence on the annealing process, and the theoretical global convergence has strict restrictions on the annealing process, resulting in poor optimization time performance. The two criteria of SA can control the convergence of the algorithm to prevent premature convergence, and the parallel sampling process can improve the algorithm's optimization time performance if the two algorithms are combined.

PSOSA hybrid algorithm combines the characteristics of PSO and SA in optimization mechanism, structure, and operation, and gives full play to the good search ability of particle swarm optimization and the characteristics of simulated annealing algorithm to effectively avoid falling into local optimization and eventually tend toward global optimization, allowing PSO and SA to learn from each other and enhance the algorithm's optimization performance.

4.3 The Introduction of Entropy

Information theory uses information entropy to quantify the quantity of information. The information entropy of a system is lower the more organized it is. In contrast, the greater the chaos of a system, the greater its information entropy [7-9]. Consequently, information entropy can also be considered a measure of system ordering.

Here, we regard the population as a system, and the level of individual differences in a population is measured by the data entropy of individual factors in the population, and the w value of the particle swarm optimization algorithm is controlled by the population information entropy. In the particle swarm optimization algorithm, when the value range of w is large, it is beneficial to search the solution space in a large range, but in this case, the search accuracy is often not high because the search is too rough, while the search accuracy will be improved when the value range of w is small, but the algorithm may fall into local extremum. Therefore, in this paper, the adaptive transformation of w is carried out in the form of formula (5), so that it is gradually reduced in the process of search. At the beginning of iterative search, w is large enough to make the algorithm have strong global retrieval ability in a larger retrieval space. With the increasing frequency of iterations, the value of w decreases adaptively to improve the accuracy of the optimal solution searched by the algorithm.

Therefore, the inertia parameter w is adjusted in the following way: suppose there are M individuals in the population, each individual is composed of $2n$ genes. Suppose R_j is the combination of the gene values j of M individuals., b_{ij} represents the number of repeats of the j gene value of the I individual in R_j , and

P_{ij} represents the share of j gene value of individual I in R_j , so the data entropy of j gene in the cluster can be expanded by the following expression[10]:

$$H_j = \sum_{i=1}^M \frac{P_{ij}}{b_{ij}} \log_2 \frac{1}{P_{ij}} \quad (3)$$

The diversity of a group can be explained by the average information entropy h of all individuals in the group:

$$H = \frac{1}{2n} \sum_{j=1}^n \sum_{i=1}^M \frac{P_{ij}}{b_{ij}} \log_2 \frac{1}{P_{ij}} \quad (4)$$

The adaptive inertia parameter w is:

$$w = \frac{w_2 H + w_1 (\log_2 M - H)}{\log_2 M} \quad (5)$$

Where, w_1 is the maximum allowable coefficient of inertia and w_2 is the minimum allowable coefficient of inertia.

4.4 Hybrid Particle Swarm Optimization Based on Entropy

The combination of particle swarm improvement (PSO) and simulated annealing (SA) can effectively deal with the allocation problem of flexible operation workshop. The measures taken are as follows:

(1) Set the inertia coefficient, acceleration constant, population number and annealing rate, initial temperature and termination temperature for the simulated annealing algorithm of particle swarm optimization, as well as initialize the position vector and velocity vector with a random value.

(2) for each particle, the velocity vectors $\mathbf{V}_{\text{process}}[L], \mathbf{V}_{\text{machine}}[L]$ are calculated by formula (1), the position vectors $\mathbf{X}_{\text{process}}[L], \mathbf{X}_{\text{machine}}[L]$ are calculated by formula (2), and the population entropy of this generation is calculated according to the population entropy formula.

(3) After decoding each particle, the applicability function of each particle (reciprocal of processing time) is calculated to find the best solution $\mathbf{P}_i$ for each particle and the best solution $\mathbf{P}_g$ for the whole group.

(4) If a particle's fitness value is greater than the fitness value of its historical optimal position, then the particle's current fitness is its optimal fitness and its current site is the optimal site in the whole world.

(5) Find the optimal solution of this generation population and determine whether the best solution $\mathbf{P}_g$ is better than the whole group. If it's better than the best solution

for the whole group update the optimal solution of the whole population with the optimal solution of this generation population.

(6) judge whether the Metropolis criterion is satisfied, if so, perform simulated annealing operation $T(t+1) = \lambda T(t)$ on the position vectors $\mathbf{X}_{\text{process}}[L]$ and $\mathbf{X}_{\text{machine}}[L]$ of the optimal solution $\mathbf{P}_g$ respectively, and store the current optimal state, turn (9), if not, turn (7).

(7) generate a new state from the current stable state, accept the new state with probability $\min [1, \exp(-\Delta t'/T)]$, and return (6).

(8) update the inertia coefficient w with formula (5) according to the population entropy calculated in step (2).

(9) Determine whether the iteration frequency exceeds the specified value, and if it is lower than the specified value, move to step (2).

(10) After the optimization, the optimal solution is output.

5. Case Analysis

Taking the data of F46(4×6), F88(8×8), F1010(10×10), MK07(20×5) problem as example, the optimization calculation was carried out for 10 consecutive times, in which the population number was 50 and the evolutionary algebra was 30, $\omega_{\max} = 1.2$, $\omega_{\min} = 0.2$, $c_1 = c_2 = 2.0$, $\lambda = 0.9$. The results of the ten ongoing operations are shown in Table 1 below. The results of simulation data prove that the improved method of composite particle swarm in this paper can always find the best solution, and that its optimization effect is clearly superior to that of the standard particle swarm optimization algorithm, despite its longer optimization time.

Table 1. Optimization result of samples

example	algorithm	time	minimum	Maximum	average
F46	PSO	0.5	17	18	17.4
	this text	13.8	17	17	17
F88	PSO	1.4	24	28	26.9
	this text	2.8	14	15	14.1
F1010	PSO	3.9	20	26	23.8
	this text	183.5	7	9	7.5
MK07	PSO	29.4	197	218	207.8
	this text	1501.6	141	155	149.6

6. Concluding Remarks

For large scale flexible operation shop allocation problem, a kind of composite particle swarm algorithm for flexible operation shop improvement allocation is presented by combining particle swarm algorithm, simulated annealing algorithm and population

entropy. At the same time, an example is used to simulate it. The simulation results show that this algorithm can effectively deal with flexible workshop allocation problems of various scales, and it has obvious advantages compared with the traditional optimization algorithm. The algorithm makes a useful exploration on the research of flexible shop scheduling optimization and provides a useful reference for the application of hybrid intelligent optimization algorithm in scheduling problems.

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